Malik Mir Hamza Nayab

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PROFESSIONAL SUMMARY

Computer Science graduate (CGPA 3.72/4.0) and first-author researcher in low-resource and multilingual Natural Language Processing and adaptive algorithms. Released a public Roman Urdu emotion dataset (Zenodo), benchmarked Arabic and multilingual transformers using bootstrap confidence intervals and McNemar tests, and designed LadderSort, a stable sorting algorithm with a proven optimality result that runs up to 1.45x faster than std::sort and TimSort on real-world data. Seeking fully funded graduate study in NLP and Machine Learning.

Research Interests: Low-resource and multilingual NLP, dialectal robustness, emotion and sentiment analysis, retrieval-augmented generation, adaptive and parallel algorithms.

EDUCATION

COMSATS University Islamabad, Abbottabad Campus — **Abbottabad, Pakistan**
Bachelor of Science in Computer Science, CGPA 3.72/4.0 — *Spring 2026*

- Thesis: Multilingual mental-health chatbot using Retrieval-Augmented Generation (RAG) and custom bilingual (English and Roman Urdu) datasets. Advisor: Ms. Ghanwa Batool.
- Relevant coursework: Data Structures and Algorithms, Machine Learning, Statistics, Parallel and Distributed Computing, Operating Systems, Database Management Systems, Compiler Construction, Human-Computer Interaction.

Harvard University (Online)
CS50 Bootcamp — *2023*

- Completed 10 problem sets, 9 labs, and a full-stack final project.

Edwardes College Peshawar — **Peshawar, Pakistan**
Salutatorian — *2020*

- Founding Coordinator of the STEM Society; Student Expo Ambassador; Student Welfare Member.

RESEARCH EXPERIENCE

LadderSort: Adaptive Stable Sorting for Partially Ordered Data — **First Author, Advisor: Dr. Osman Khalid**

- Designed and implemented in C++ a deterministic, comparison-based stable sorting algorithm that adapts to K, the minimum number of nondecreasing subsequences; proved optimality when K = LDS(A).
- Built a two-phase architecture: hinted greedy decomposition with exponential galloping, followed by a dynamically selected K-way loser-tree tournament or two-way galloping merge.
- Benchmarked against std::sort and TimSort on Apple M1 with up to 100M elements: up to 1.45x speedup on GitHub Archive event streams (N = 55K, K = 368) and 30–50% gains on synthetic interleaved workloads.

Cross-Dialect Robustness of Arabic Pretrained Language Models for Sentiment Analysis — **First Author**

- Built a multi-dialect evaluation framework on the ArSarcasm corpus (5,207 tweets) comparing AraBERT, MARBERT, and CAMeL-BERT across five varieties: Egyptian, Gulf, Levantine, Maghrebi, and Modern Standard Arabic.
- Proposed the variance-based Dialect Robustness Index (DRI); validated results with 1,000-resample bootstrap confidence intervals and paired McNemar tests ($p < 0.001$).
- Diagnosed and fixed a memory artifact in zero-shot checkpoint configurations, showing that fine-tuned task adaptation significantly outperforms frozen representations.

Roman Urdu Emotion Classification Benchmark (GoEmotions) — **First Author**

- Evaluated mBERT, XLM-R, and MuRIL on 28-class Roman Urdu emotion classification with frozen splits; XLM-R achieved the best Macro-F1 (0.2909).
- Ran ablations on spelling-variant augmentation (k = 0, 1, 3) and training length (2–4 epochs) with strict data-leakage controls and matched hyperparameters across models.
- Established zero-shot baselines by mapping GoEmotions to Ekman-6 (70.31% accuracy, Macro-F1 0.5004); produced reports and confusion matrices for reproducibility.

Roman Urdu GoEmotions Dataset — **Zenodo DOI: 10.5281/zenodo.18445910**

- Built a fault-tolerant English-to-Roman Urdu translation pipeline using Gemma-3 (4.3B) via Ollama to create a large public emotion dataset for low-resource NLP.
- Implemented resumable 100-row batch processing with emotion-preserving constraints, automated quality control, and anomaly flagging, designed to run within Kaggle and Colab session limits.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, TypeScript, SQL, C#, Dart

Machine Learning and NLP: PyTorch, Hugging Face Transformers, RAG pipelines, Scikit-learn, NumPy, Pandas

Research Methods: Bootstrap confidence intervals, McNemar test, Welch t-test, ablation design, reproducible benchmarking, data visualization

Parallel and Systems: CUDA C, OpenMP, algorithm design and analysis, operating systems

Web, Cloud, Databases: React, Node.js (Express), Flask, Flutter, PostgreSQL, MongoDB, Vercel, Heroku, Cloudflare R2, GitHub Actions

PROJECTS

Multilingual Mental Health Chatbot (Undergraduate Thesis) **Gemma-3, RAG, React, Flask, MongoDB**

- Designed and deployed a bilingual (English and Roman Urdu) conversational system combining Gemma-3 (1B) with RAG over vector embeddings.
- Trained PyTorch emotion classifiers: 86% test accuracy (BERT on DAIR) and approx. 90% (XLM-RoBERTa on a custom Ekman-6 dataset); deployed inference on Modal serverless GPUs.

MiddelMen (TrustThread): middelmen.com **React, Node.js, TypeScript, PostgreSQL**

- Sole developer of a live social-commerce trust platform: React/Vite front end on Vercel, Node.js/Express API on Heroku, and CI with GitHub Actions.
- Designed a PostgreSQL schema (Drizzle ORM) with role-based access control, verified seller reviews, and dispute logging; implemented Google OAuth 2.0 and bcrypt authentication.
- Added crawler-aware server-side rendering with Schema.org JSON-LD and OpenGraph metadata for SEO; used Cloudflare R2 for media storage.

GPU/CPU Object Detection Engine **C, CUDA C, OpenMP**

- Implemented BMP image processing with grayscale conversion and Sobel edge detection, then parallelized it across CPU threads (OpenMP) and GPU kernels (CUDA C) with memory-access optimization.

LEADERSHIP AND COMMUNITY

Google Developer Groups on Campus (GDGoC), COMSATS Abbottabad **Abbottabad, Pakistan**

Interim Lead and Co-Lead *Fall 2025 – Fall 2026*

- Led multi-track technical bootcamps (AI/ML, web architecture, Flutter, game development) and managed logistics and judging for the university-wide Final Year Project Competition.
- Built partnerships with BASE Chain Pakistan, Systems Limited, Fiverr, and Thrive Pakistan; hosted industry masterclasses and co-organized COMBITS (Wah Campus) and Hazara Tech Fiesta 2025.

Google Developer Groups on Campus (GDGoC), COMSATS Abbottabad **Abbottabad, Pakistan**

Management Lead *Spring 2024 – Spring 2025*

- Organized DevFest Pakistan 2024 (Abbottabad Edition) and led the COMSATS delegation to GDG Cloud DevFest 2024 at NIC Islamabad.
- Moderated 4 parallel bootcamps of 15+ hands-on sessions each; ran membership drives that raised student retention; pitched TrustThread at the VisioSpark Startup Idea Competition.

WORK EXPERIENCE

Monet Real Estate **Dubai, UAE**

Off-Plan Property Specialist *2024*

- Researched Dubai off-plan market trends and pricing to advise high-net-worth clients.
- Negotiated between international buyers, developers, and sellers, generating repeat business.

Enchant Marketing **Peshawar, Pakistan**

Founder and Digital Marketing Lead *Sep 2019 – Sep 2020*

- Founded a digital marketing agency and managed a network of 50+ bloggers, creators, and micro-influencers.
- Led a Decipherer Bookstore Instagram campaign: approx. 200K impressions, approx. 50% higher direct sales.

HONORS AND AWARDS

- Leadership Appreciation Award (2025), COMSATS University Islamabad and Google Developer Groups, for contributions to regional technical education.
- Salutatorian, Edwardes College Peshawar (2020).
- Doubles Runner-up, Inter-University Table Tennis Championship.